

Side-Channel Analysis Revisited and Evaluated

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Abstract. Side-channel analysis recovers a secret by exploiting the key-dependent characteristics of the leakages. Practical techniques, such as Distance-of-Means analysis (DoM), Kolmogorov-Smirnov analysis (KSA) and Cramér-von Mises analysis (CvMA), provide valuable insights about the secret from the indirect perspectives of statistical moment and cumulative distribution function (CDF) respectively, circumventing the direct and costly estimation of leakage probability densities and therefore enabling wider applicability in practice. Though both the perspectives are informative, their relationships in the context of side-channel analysis remain unclear. In other words, the fundamental questions of "which one is better?" and "why and under what circumstances?" leave as open problems. In this paper, we introduce the probability-probability (PP) plot in statistics as a common framework for explaining the mathematical foundations of CDF-based techniques, which facilitates an intuitive understanding of different variant strategies. Then, inspired by the growth pattern of the PP curve, we propose a novel distinguisher based on the famous Mann-Kendall test, where measurements are managed with ordinality and nominality. This goodness-of-fit test checks whether a key-dependent binary sequence originates from a random binomial distribution, by efficiently searching potential label clusters. Finally, we explore the symmetry and dual counterpart of CDF in mathematics, introducing the quantile-quantile (QQ) plot and develop an interesting technique based on the inverse cumulative distribution function (ICDF). We present a general discussion of its bridging role, regarding detail capture as well as signal-to-noise ratio (SNR). On this basis, we establish the relationships among moment-based, ICDF-based, and CDF-based techniques, which additionally allows for bounding the security level of the CDF-based techniques using well-established metrics that are originally proposed for evaluating the traditional moment-based family. Experiments across various settings validate our theoretical findings and demonstrate the effectiveness of the two proposed distinguishers.

Keywords: side-channel analysis · security evaluation · PP plot · Mann-Kendall test · QQ plot.

1 Introduction

Cryptographic algorithms are designed assuming that they will be implemented in a closed environment which do not leak any information about the internal states. However, this long-standing security assumption is nullified whenever an adversary is enhanced with side-channel analyses. Physical observable phenomena (i.e., side-channel leakages), such as power consumption [17] [7], are generated inadvertently during the computation and statistically depend on the sensitive values being processed. If properly exploited, these leakages provide a breach for thwarting the protection. Over the past decades, side-channel analyses have experienced a rich development and become a non-negligible threat to today’s cryptosystems.

On the whole, side-channel analysis captures dependency between the secret and the leakages through a leakage model. From coarse to fine, there are plenty of choices for the model (see the discussion in [37]), such as the generic identity model [4], the well-established a-priori Hamming weight (HW) and Hamming distance (HD) models, and the sophisticated profiled model based on linear regression [27]. Analogously, on the other hand, leakages can also be exploited at different levels of granularity: directly through probability densities, or indirectly via distribution characteristics.

The density-aware analysis aims to utilize full information in leakage distribution and explore the “worst case” that cryptosystem can encounter. They are able to remain efficient at all times when a precise estimation of probability densities is allowed, especially in cases where the simple Gaussian template is no longer effective. Naturally, the estimation process incurs additional costs in time, computation, and resources. Also, the concrete methods for estimation (e.g., parameterized Gaussian template [8] or non-parameterized histogram [24]) should be selected and validated as appropriate through leakage certification (e.g., divergence based [11] and moment based [10] certification), by assessing statistical unbiasedness. Representatives of the density-aware analyses include maximum likelihood attack (ML-attack) [8], mutual information analysis (MIA) [32], χ^2 analysis [24] and so on.

In contrast, the characteristic-aware analysis prioritizes practical usability, and develops solutions that can always come in handy for “common cases”. To attain this goal, this type of technique is completely free of parameters and simply partitions leakages according to a coarse-grained model (note that some density-aware methods like MIA also need this partition step), such as the identity model and the HW model. The generic statistical test then captures the secret dependency from a distribution characteristic perspective (e.g., statistical moment or cumulative distribution function), since it only needs to tell whether the partition is meaningful (i.e., a clear distinction shows up) rather than resorting to some dedicated and explainable metric [29] (e.g., the likelihood value in ML-attack and

the shared entropy in MIA). Representatives of the characteristic-aware analysis include Distance-of-Means analysis (DoM) [17], Kolmogorov-Smirnov analysis (KSA) [36], Cramér-von Mises analysis (CvMA) [32], and so on.

Interestingly, there are some other techniques searching for a *trade-off* between performance and practicality. For some particular cases where leakages are approximatively linear, the specific linear correlation coefficient is usually a good choice. For instance, the Pearson’s coefficient [7] is suitable for proportional a-priori models whilst the Spearman’s coefficient [3] is preferable for ordinal ones. Note that the recent Gini-impurity index analysis [28] should also belong to this type because it is essentially a combination of the Pearson’s and Spearman’s coefficients [38].

So far, there has been a great deal of research devoted to the density-aware analysis. Moreover, the increased variability in today’s modern nanoscale technology has progressively invalidated the portability and reusability of the estimated densities [33] [25], giving rise to new problems such as domain adaptation (i.e., a noticeable gap emerges between leakage distributions of different device). Therefore, shifting the research focus to the more practical characteristic-aware analysis, including introducing novel distinguishers and exploring relationships among most commonly-used techniques, appear to be motivating and challenging.

1.1 Related Works

Various practical techniques have been proposed. Linear regression analysis (LRA) [27] is an influential one due to its unique “on-the-fly” profiling mechanism [9]. However, in recent works [19], the regression coefficients are proved to be equivalent to DoM’s distinguishing values, with the corresponding explanatory variables serving as the selection functions. Given that the general non-device-specific intuition advised in [37] solely applies to the regression coefficients, it will be sufficient to focus on the basic DoM.

Another remarkable technique is copulas [33]. But, as the authors point out themselves, the incorporated distance sampling demonstrates advantages mainly for high-dimensional leakages. More importantly, it is not rigorously free of parameters, as it needs to select a smoothing function (e.g., the Epanechnikov kernel in [33]) together with a window size parameter for noise filtering. This could again lead us to the tricky situation where the concrete setting employed plays a decisive role (e.g., see the experiments of MIA in [32] [4]).

In this paper, we concentrate on the most widely used and minimally constrained techniques: KSA, CvMA and DoM (note that we also consider their typical variants). DoM and its variants handle leakages from the statistical moment’s aspect, while KSA and CvMA start from the cumulative distribution function. Though both are limited compared to the full distribution in the density-aware techniques, they provide valuable insights and have undergone a long development.

Moment-based techniques. DoM targeting a single bit of the sensitive variable (refer to as SB-DoM) is the first proposed side-channel technique [17].

It is soon extended to multiple bits: the generalized DoM (G-DoM) [5], the all-or-nothing DoM (AON-DoM) [23], and the robust DoM [9], in order to overcome some algebraic property that leads to failure in the mono-bit setting (e.g., the see “ghost peaks” in [7]). Among them, AON-DoM is less favored because of its waste of leakage measurements [6]. Showing strong practicality, this moment-based technique is consolidated in the test vector leakage assessment (TVLA) [13] whose derivative becomes a part of the NIST famous standard FIPS 140.

CDF-based techniques. Originally suggested in [32], KSA and CvMA are promising competitors to MIA. Though recognized as suboptimal in [36], later study [39] constructs variants to make KSA superior in some circumstances. Among them, it is noteworthy that some follow Gaussian assumption and base on the so-called partial KSA (PKSA) [16]. These variants of low generality are not in consideration. We keep both the cumulative and non-cumulative partition strategy for variant construction, and discard the normalization because it does not always yield a profit [39]. Inter-class KSA that compares conditional distributions between themselves instead of comparing them with global distribution are presented in [20]. In recent works [40], KSA is applied for efficient leakage detection, just like DoM in TVLA. Analogous to KSA built on maximum deviation, CvMA examining squared deviations is often regarded as a twin brother and discussed together (e.g., see [31]). The variant strategies of KSA can be directly adopted to CvMA without any modification.

Although the perspectives of statistical moment and CDF can both provide informative insights about the secret, their relationships still remain unclear. In other words, the fundamental questions of “which is better?” and “why and under what circumstances?” leave as open. In [15], the authors make an initial attempt towards the evaluation of KSA and DoM, assuming a single-bit intermediate variable – as they state that “it is very much intricate in the multi-bit case” – as well as a simple unimodal distribution, which they themselves acknowledge to be valid only in low-SNR scenarios with binomial algorithmic noise. The situation is even less clear for CvMA in that it is hardly investigated. Therefore, we believe that getting rid of the restrictive assumptions, deriving an intuitive framework and theoretically exploring the relationships in a general manner (including the multi-bit setting) will bring new perspectives to the side-channel community.

1.2 Contributions

In this paper, our contributions are as follows:

- We take an in-depth look into the misalignment problem of leakage measurements which occurs during the calculation of KS and CvM divergence in practice. We ground these CDF-based techniques in the theoretical framework of the probability-probability (PP) plot, where KS and CvM divergence are mapped to the maximum and total squared distances, respectively, between a continuous reference line and a discrete curve. This framework enables intuitive comparisons of different variants and captures their intrinsic differences by means of margin analysis.

- Inspired by the growth pattern in PP plot, we introduce ordinality and nominality simultaneously to manage leakage measurements. Different from previous works, the task of identifying meaningful partitions is refined by a goodness-of-fit test checking whether a key-dependent binary sequence is an instance of a binomial distribution with known parameters. The novel Mann-Kendall test in statistics is for the first time applied to assist side-channel key recovery, by efficiently searching potential clusters of nominal labels. Experiments across various settings confirm that this enhanced CDF-based scheme is a significantly better alternative to all variants of KSA and CvMA, and is close to optimality.
- Finally, we extend the theoretical foundation of the PP plot by turning to its symmetry and dual counterpart, which leads to the quantile-quantile (QQ) plot and an interesting family of ICDF-based technique. We present a general discussion about their equivalence in the context of side-channel analysis, regarding detail capture and SNR. On this basis, we establish the relationships between the CDF-based and moment-based techniques. This connection allows us to bound the security levels of the CDF-based techniques using well-established metrics (e.g., the confusion coefficient) that are originally proposed for evaluating the traditional moment-based techniques.

1.3 Organization

The remaining of this paper is organized as follows: preliminaries including leakage model, variants of distinguishers, and evaluation metrics are introduced in Section 2. The PP plot and comparisons of techniques are described in Section 3. We propose the Mann-Kendall test as an improvement with advanced ordinality and nominality in Section 4. The QQ plot and relationships of distinguishers are derived in Section 5. Experiments on simulated and real leakages of PRESENT and AES on software and hardware implementations are given in Section 6. Finally, we conclude this paper in Section 7.

2 Preliminaries

2.1 Side-channel Leakages and Leakage Model

Side-channel analysis treats the secret as a tuple of subkeys and recover them separately. Let k^* denote the target subkey selected from a set $\mathcal{K}$ and k denote any possible guessing candidate. Let m denote the plaintext byte, $\mathcal{H}$ denote the linear operation (e.g., bitwise XOR), and $\mathcal{G}$ denote the sensitive non-linear operation (e.g., S-box) whose output is an n -bit intermediate value that can be exploited for secret recovery. Let $\mathcal{F}$ denote the leakage function describing the physical signals leaked according to the computation of $\mathcal{G}$ (e.g., HW). Then, the leakage measurement $\mathcal{L}$ can be written as:

$$\mathcal{L} = \mathcal{F} \circ \mathcal{G} \circ \mathcal{H}(m, k^*) + \mathbf{N}_m = \varphi(m, k^*) + \mathbf{N}_m. \quad (1)$$

$\mathbf{N}_m$ is the Gaussian noise with a zero mean and a variance of $\mathbb{D}\{\mathbf{N}_m\} = \sigma_m^2$. The overall distribution of side-channel leakages over the fixed secret subkey k^* and random plaintext byte m is a convex combination of 2^n Gaussian distributions (i.e., Gaussian mixture). Compared to previous works, our leakage model is more general and practical in two aspects:

(1) We do not assume homogeneity of the noise. The variance of noise corresponding to different calculations can be different from each other, i.e., $\sigma_0^2 \neq \sigma_1^2 \neq \dots \neq \sigma_{2^n-1}^2$.

(2) We do not assume a Gaussian overall distribution (e.g., as in PKSA [16] [39]). While the Gaussian mixture may approach a Gaussian distribution under a low SNR, it generally captures the intuition and details of real-world scenarios where leakage distributions are unknown and can be highly diverse.

2.2 Distance-of-Means Analysis

DoM partitions leakage measurements according to a binary select function. Let $L = \{\mathcal{L}_1, \mathcal{L}_2, \dots, \mathcal{L}_S\}$ denote the universal leakage measurement set sorted in ascending order, and $L_1 = \{L | \theta \in \Omega_1\}$ and $L_2 = \{L | \theta \in \Omega_2\}$ denote the two subsets after the partition. The notation θ is the select function and $\Omega_1, \Omega_2 \subseteq \text{Range}(\theta)$ (i.e., the range of the function). Leakages bundled together are deemed to share the same distribution. Differentials of means are calculated across partitions to select subkey candidates:

$$\begin{aligned} \mathbf{D}_{\text{DoM}} &= \arg \max_{k \in \mathcal{K}} \Delta_{\text{DoM}}^{(k)}(t) \\ &= \arg \max_{k \in \mathcal{K}} \hat{\mathbb{E}}\{L_1\} - \hat{\mathbb{E}}\{L_2\}. \end{aligned} \quad (2)$$

For SB-DoM [17], the select function is the guessing value of one target bit, i.e., $L_1 = \{L | x^{[t]} = 0\}$ and $L_2 = \{L | x^{[t]} = 1\}$, in which x is the n -bit intermediate value calculated under guessing subkey k and $x^{[t]}$ is the t -th bit (i.e., the target). Usually, for software implementations, x is chosen as $x = \mathcal{G} \circ \mathcal{H}(m, k)$, while for hardware implementations it becomes $x = (\mathcal{G} \circ \mathcal{H}(m, k)) \oplus \mathcal{H}(m, k)$. SB-DoM follows a non-cumulative partition strategy where each possible value of θ is assigned to a distinct partition. It is soon extended to multiple bits by incorporating the cumulative partition strategy in G-DoM [5], i.e., $L_1 = \{L | x < \lfloor \frac{n}{2} \rfloor\}$ and $L_2 = \{L | x > \lceil \frac{n}{2} \rceil\}$. In this cumulative case, several values of θ are assigned to one partition. For clarity, we use $\Delta_{\text{SB-DoM}}$ and $\Delta_{\text{G-DoM}}$ to distinguish between the differential values.

Robust extensions of DoM are given in [9] to enhance the reliability, in the event of negative peaks:

$$\mathbf{D}_{\text{DoM}} = \arg \max_{k \in \mathcal{K}} (\Delta_{\text{DoM}}^{(k)})^2, \quad \Delta_{\text{DoM}}^{(k)} = \begin{cases} \Delta_{\text{SB-DoM}}^{(k)}, \\ \Delta_{\text{G-DoM}}^{(k)}. \end{cases} \quad (3)$$

This extension has no impact on the attack performance.

2.3 Kolmogorov-Smirnov and Cramér-von Mises Analysis

KSA and CvMA can be conducted under binary or multiclass partitioning schemes. In this paper, to ensure their comparability with DoM, which inherently relies on a binary partition of measurements, we focus on the binary versions of KSA and CvMA, while still considering all their inter-class, intra-class, cumulative and non-cumulative variants.

KSA and CvMA are typical non-parametric tools for quantifying differences between two distributions. They define an empirical CDF on the measurement set $Z = \{z_1, \dots, z_N\}$ as:

$$\hat{F}_Z(z_t) = \frac{1}{N} \sum_{i=1}^N \chi_{z_i}, \quad \chi_{z_i} = \begin{cases} 1 & \text{if } z_i \leq z_t, \\ 0 & \text{otherwise.} \end{cases} \quad (4)$$

The set Z can be either L , L_1 or L_2 . By the Glivenko-Cantelli theorem [14], the empirical CDF converges almost surely towards the theoretical CDF, uniformly over all z [33]:

$$\|\hat{F}_Z - F_Z\|_\infty \equiv \sup_{z \in \mathbb{R}} |\hat{F}_Z(z) - F_Z(z)| \xrightarrow{a.s.} 0. \quad (5)$$

This supports the soundness of KSA and CvMA in practical scenarios of limited measurements. KSA and CvMA identify the secret subkey based on the fact that the correct candidate will lead to significantly different subsets of the partition. Intra-class KSA (iKSA) measures the maximum deviations between the conditional distributions and the global distribution, while inter-class variant (eKSA) compares the conditional distributions between themselves:

$$\begin{aligned} \mathbf{D}_{\text{iKSA}} = \arg \max_{k \in \mathcal{K}} & \left(\frac{1}{2} \sup_{z \in \mathbb{R}} |\hat{F}_L(z) - \hat{F}_{L_1}(z)| \right. \\ & \left. + \frac{1}{2} \sup_{z \in \mathbb{R}} |\hat{F}_L(z) - \hat{F}_{L_2}(z)| \right). \end{aligned} \quad (6)$$

$$\mathbf{D}_{\text{eKSA}} = \arg \max_{k \in \mathcal{K}} \sup_{z \in \mathbb{R}} |\hat{F}_{L_1}(z) - \hat{F}_{L_2}(z)|. \quad (7)$$

All of them can be conducted in a non-cumulative manner as SB-DoM with $L_1 = \{L | x^{[t]} = 0\}$ and $L_2 = \{L | x^{[t]} = 1\}$, or in a cumulative manner as G-DoM with $L_1 = \{L | x < \lfloor \frac{n}{2} \rfloor\}$ and $L_2 = \{L | x > \lfloor \frac{n}{2} \rfloor\}$. The corresponding distinguishers are denoted as SB-iKSA, G-iKSA, SB-eKSA and G-eKSA respectively. In contrast, the twin brother CvMA evaluates the squared distances for key discrimination:

$$\begin{aligned} \mathbf{D}_{\text{CvMA}} = \arg \max_{k \in \mathcal{K}} & \left(\frac{1}{2} \int_{-\infty}^{+\infty} (\hat{F}_L(z) - \hat{F}_{L_1}(z))^2 dz \right. \\ & \left. + \frac{1}{2} \int_{-\infty}^{+\infty} (\hat{F}_L(z) - \hat{F}_{L_2}(z))^2 dz \right). \end{aligned} \quad (8)$$

$$\mathbf{D}_{\text{eCvMA}} = \arg \max_{k \in \mathcal{K}} \int_{-\infty}^{+\infty} (\hat{F}_{L_1}(z) - \hat{F}_{L_2}(z))^2 dz. \quad (9)$$

Similarly, the variants following the non-cumulative and cumulative strategies are denoted as SB-iCvMA, G-iCvMA, SB-eCvMA, and G-eCvMA accordingly.

2.4 Summary of Distinguisher Variants

We use $\mathbf{D}(P||Q)$ to specify the two distributions P, Q to compare in the distinguisher $\mathbf{D}$, and provide the following table as a clear summary of all the variants we concern in this paper. We reformulate iKSA and iCvMA in the form of eKSA and eCvMA. Robust extensions of DoM are not included.

Table 1: Summary of DoM, KSA and CvMA.

$\mathbf{D}_{\text{SB-DoM}} = \mathbf{D}_{\text{DoM}}(\{L x^{[t]} = 0\} \{L x^{[t]} = 1\})$
$\mathbf{D}_{\text{G-DoM}} = \mathbf{D}_{\text{DoM}}(\{L x < \lfloor \frac{n}{2} \rfloor\} \{L x > \lceil \frac{n}{2} \rceil\})$
$\mathbf{D}_{\text{SB-iKSA}} = \frac{1}{2}\mathbf{D}_{\text{eKSA}}(L \{L x^{[t]} = 0\}) + \frac{1}{2}\mathbf{D}_{\text{eKSA}}(L \{L x^{[t]} = 1\})$
$\mathbf{D}_{\text{SB-eKSA}} = \mathbf{D}_{\text{eKSA}}(\{L x^{[t]} = 0\} \{L x^{[t]} = 1\})$
$\mathbf{D}_{\text{G-iKSA}} = \frac{1}{2}\mathbf{D}_{\text{eKSA}}(L \{L x < \lfloor \frac{n}{2} \rfloor\}) + \frac{1}{2}\mathbf{D}_{\text{eKSA}}(L \{L x > \lceil \frac{n}{2} \rceil\})$
$\mathbf{D}_{\text{G-eKSA}} = \mathbf{D}_{\text{eKSA}}(\{L x < \lfloor \frac{n}{2} \rfloor\} \{L x > \lceil \frac{n}{2} \rceil\})$
$\mathbf{D}_{\text{SB-iCvMA}} = \frac{1}{2}\mathbf{D}_{\text{eCvMA}}(L \{L x^{[t]} = 0\}) + \frac{1}{2}\mathbf{D}_{\text{eCvMA}}(L \{L x^{[t]} = 1\})$
$\mathbf{D}_{\text{SB-eCvMA}} = \mathbf{D}_{\text{eCvMA}}(\{L x^{[t]} = 0\} \{L x^{[t]} = 1\})$
$\mathbf{D}_{\text{G-iCvMA}} = \frac{1}{2}\mathbf{D}_{\text{eCvMA}}(L \{L x < \lfloor \frac{n}{2} \rfloor\}) + \frac{1}{2}\mathbf{D}_{\text{eCvMA}}(L \{L x > \lceil \frac{n}{2} \rceil\})$
$\mathbf{D}_{\text{G-eCvMA}} = \mathbf{D}_{\text{eCvMA}}(\{L x < \lfloor \frac{n}{2} \rfloor\} \{L x > \lceil \frac{n}{2} \rceil\})$

2.5 Evaluation Metrics

We recall a few evaluation metrics for side-channel analysis.

Confusion coefficient. Cryptographic algorithms are designed with confusion to resist against cryptanalysis. Confusion obscures the statistical correlation between the ciphertext and the secret key. Proposed in [12] and generalized in [18], confusion coefficient adapts the concept of confusion to the field of side-channel analysis. For a pair of subkey candidates (k, k^*) , the outputs of $\varphi(m, k)$ and $\varphi(m, k^*)$ differ over the same plaintext byte variable m . The degree of deviation indicates the efforts required to distinguish them using side-channel leakages. For this purpose, the confusion coefficient is defined as the mathematical

expectation of squared distance:

$$\kappa(k, k^*) = \mathbb{E}\{(\varphi(m, k) - \varphi(m, k^*))^2\}. \quad (10)$$

This metric has been adopted as a powerful tool for quantifying and explaining the attack efficiency of the DoM family.

Security margin. Security margin suggested in [34] [35] has been shown to be more expressive and informative than the success rate [34] [26]. Typically, a side-channel distinguisher computes a distinguishing value for each guessing subkey, and returns the candidate corresponding to the maximum value as the secret subkey, according to the maximum likelihood principle. The security margin is defined as the difference between the distinguishing values under the correct subkey and the incorrect candidates. Specifically, the standard margin is expressed as:

$$\frac{\mathbf{D}[k^*] - \mathbb{E}\{\mathbf{D}[k] | k \in \mathcal{K}\}}{\sqrt{\mathbb{D}\{\mathbf{D}[k] | k \in \mathcal{K}\}}}, \quad (11)$$

where $\mathbf{D}[k]$ denotes the distinguishing value (i.e., the output of the distinguisher) calculated under the subkey guess k . Obviously, a larger margin generally implies better distinguishability: small distinguishing values of incorrect candidates are less likely to surpass the peak value associated with the correct subkey due to the influence of noise.

3 Foundations and Comparisons of KSA and CvMA Distinguishers

3.1 The Misalignment Problem

If we recall Equ. (6)~(9), calculating KS or CvM divergence requires finding leakage measurements with identical values, as indicated by the term z in the equations, from the two measurement sets being compared. Then, the empirical CDF values at these aligned measurements are paired and examined. However, in practice, leakages are continuous and the measurement sets are finite. This gives rise to a misalignment problem where identical measurements are absent.

Fig. 1(a) illustrates a toy example of the leakage misalignment with the sample sets $P = \{21.56, 73.89, 67.54, 89.01\}$ and $Q = \{12.01, 94.07, 8.15, 24.31\}$. The x-axis represents the actual leakage measurement values and the y-axis shows the corresponding CDF values. In this example, the KS and CvM divergence cannot be calculated on their definitions because actual CDF points do not simultaneously exist at the same x-coordinate. To overcome this issue, an interpolation method is applied to generate estimated CDF values. In general, the interpolation predicts an unknown value according to a combination of its adjacent actual values. Typical methods include linear, polynomial, and spline interpolation. Yet, their designs are tailored for specific purposes. To preserve the original features as much as possible, previous-value interpolation is most commonly used, in which the interpolated value is simply set to its nearest actual

value that is smaller than itself. Therefore, it is also known as left-continuous interpolation.

Fig. 1(b) displays the application of previous-value interpolation. Horizontal line segments are added between actual CDF values of adjacent leakage measurements, with their head aligned with the left values (i.e., the smaller previous values). This process generates estimated CDF for all leakage values within the x-range of the line segment that are absent from the actual measurement sets. Then, the KS divergence calculates the maximum deviation between the two piecewise curves while the CvM divergence quantifies the area enclosed.

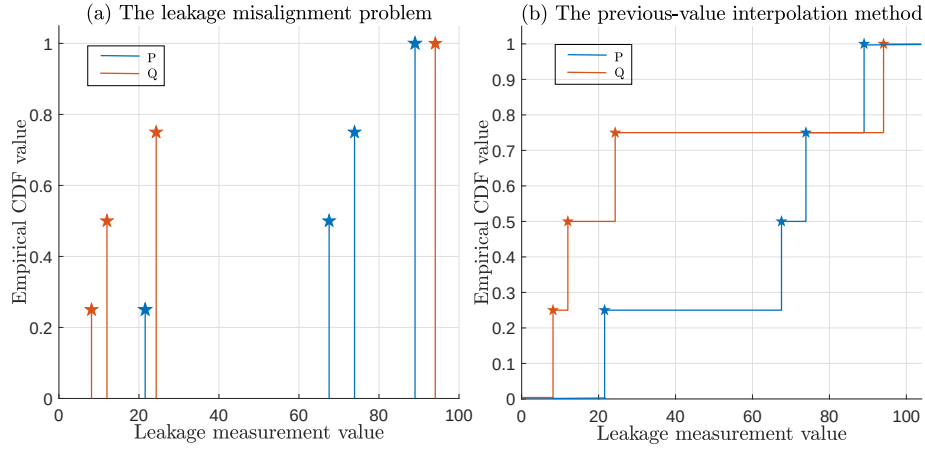


Fig. 1: The misalignment problem and the value interpolation.

3.2 Foundations of KSA and CvMA Distinguishers

In the above example, the sequence of empirical CDF value pairs to compare in the KS or CvM divergence is given by: $\{(0, 1/4), \dots, (0, 2/4), \dots, (1/4, 2/4), \dots, (1/4, 3/4), \dots, (2/4, 3/4), \dots, (3/4, 3/4), \dots, (1, 3/4), \dots, (1, 1)\}$. The explicit terms are obtained at endpoints (marked with pentagrams in the figure) of the rising edges, where at least one value in each pair is actual (i.e., not estimated). The omitted terms are CDF pairs in which both values are interpolated, such as those found in the overlapping x-ranges of the red and blue line segments. They take the same values as their nearest preceding explicit terms because of the previous-value interpolation.

As observed from the sequence, the differences between the two distributions P and Q will not be detected by the KS and CvM divergence if their empirical CDF values grow at the same rate. To make this principle intuitive, let x-axis represent the CDF values of distribution P , and y-axis represent those of distribution Q . Then, we map the sequence of empirical CDF value pairs onto a

new plot. It will exactly adhere to the straight reference line $y = x$ if and only if P and Q is identical. Otherwise, it forms a twisted curve. This graphical tool is the well-known probability-probability (PP) plot in statistics. The reference line is predefined, making it generic and independent of any specific device or algorithm.

For illustration, we offer examples of PP plots on typical HW leakages, i.e., $\mathcal{L} = \text{HW}(\text{Sbox}(m \oplus k^*)) + \text{N}$. We choose the simple non-cumulative partition as the example, with $L_1 = \{L | x^{[1]} = 0\}$ and $L_2 = \{L | x^{[1]} = 1\}$. Figs. 2~3 display the PP plots under the correct guessing subkey $k = k^*$ and a wrong candidate $k \neq k^*$. The yellow dotted line is the reference $y = x$, and it is continuous. The red, blue and pink curves are made up of discrete circles which are the mappings of empirical CDF values. In the correct subkey setting, all the discrete curves are clearly deviated from the reference, showing strong evidences that the two distributions being compared are different from each other. The blue curve (i.e., L_2 vs. L) lies above the reference. This is because the conditional distribution L_2 occupies the high HW region, because the target bit always satisfies $x^{*[1]} = 1$. So, its CDF values climb slower than that of the global distribution L in the initial stage, but becomes faster at the end. This upward distortion forms a convex shape. Similar analyses can be adopted to the other cases. On the other hand, in the setting of wrong candidate, all the curves fluctuate around the reference, implying a meaningless partition. These observations intuitively demonstrate the soundness of the CDF-based techniques for distinguishing secret subkey.

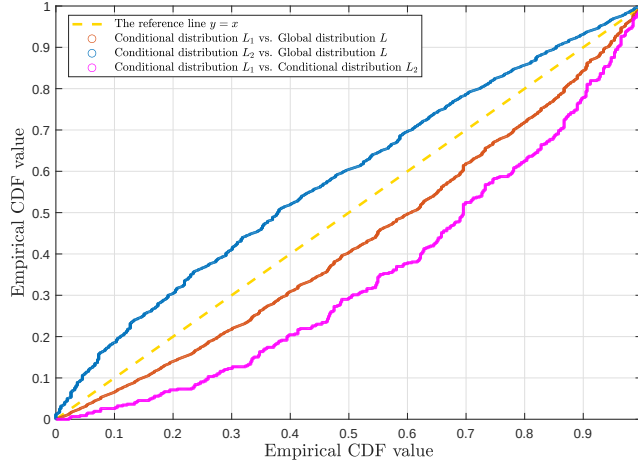
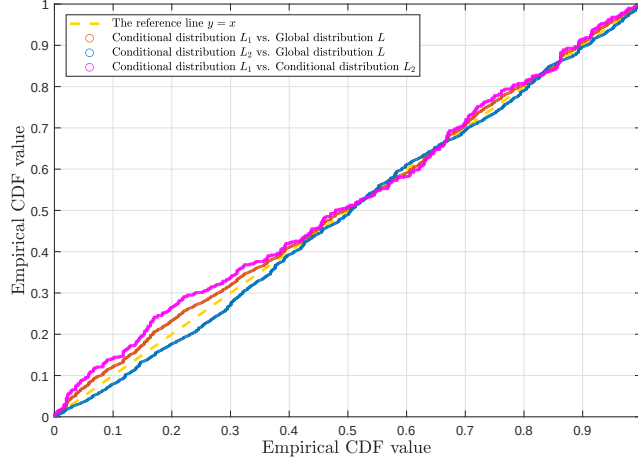


Fig. 2: PP plot under the correct guessing subkey $k = k^*$.

Fig. 3: PP plot under the wrong guessing subkey $k \neq k^*$.

3.3 Comparisons of KSA and CvMA Distinguishers

As we can conclude from Section 3.2, the deviation of the discrete curve from the reference line is the intuitive metric for distinguishing subkeys in the PP plot. Large deviations suggest distinct distributions of the subsets and the partition is meaningful. For any point (x_t, y_t) on the curve, its distance to the straight line defined by $Ax + By + C = 0$ can be obtained using the well-established geometric formula:

$$Dst = \frac{|Ax_t + By_t + C|}{\sqrt{A^2 + B^2}}. \quad (12)$$

In our case of $y = x$, it is simplified as:

$$Dst = \frac{1}{\sqrt{2}} |x_t - y_t| = \frac{1}{\sqrt{2}} |\hat{F}_P(\mathcal{L}_t) - \hat{F}_Q(\mathcal{L}_t)|, \quad (13)$$

where $\mathcal{L}_t$ is the leakage measurement value corresponding to the empirical CDF value pair (x_t, y_t) . Surprisingly, the maximum distance is equivalent to the KS's criterion:

$$\max Dst \simeq \sup_{\mathcal{L} \in \mathbb{R}} |\hat{F}_P(\mathcal{L}) - \hat{F}_Q(\mathcal{L})|. \quad (14)$$

The constant coefficient $1/\sqrt{2}$ is key-independent and thus can be ignored. Similarly, the total squared distance equals to the CvM's criterion:

$$\sum Dst^2 \simeq \int_{-\infty}^{+\infty} (\hat{F}_L(\mathcal{L}) - \hat{F}_{L_1}(\mathcal{L}))^2 d\mathcal{L}. \quad (15)$$

From these aspects, the intuitive PP plot enables straightforward comparisons between intra- and inter-class variants of KSA and CvMA through margin

analysis. Specifically, the pink curve corresponding to the inter-class variant shows significantly wider distances to the reference line than the blue and red curves, which represent intra-class variants, as depicted in Fig. 2 when $k = k^*$. However, in the case of $k \neq k^*$, the distances for both variants approach similar levels. By Equ. (11), this results in a broader security margin for the inter-class distinguishers. Consequently, when shifting from an intra-class distinguisher to its inter-class version, the distinguishing value for the correct subkey increases, meanwhile, those for the incorrect candidates remain almost unchanged. Such an increase boosts the peak value and enhances the noise resistance. This comparison framework underpins the experimental observations in [20] that inter-class variants outperform intra-class ones under usual simulated HW leakages.

4 Advancing Key Recovery: Leveraging Novel Statistical Test

4.1 The Growth Pattern of Empirical CDF Values

The growth pattern of the empirical CDF values directly determines how the discrepancy emerges between the two distributions, as reflected by the deviation of the PP curve from the reference line. In this subsection, we aim to gain a deeper understanding. For clarity, we write out the measurement subsets as $L_1 = \{p_1, p_2, \dots, p_M\}$ and $L_2 = \{q_1, q_2, \dots, q_N\}$, with the measurements sorted in ascending order. Recall the notation of the sorted universal set $L = \{\mathcal{L}_1, \mathcal{L}_2, \dots, \mathcal{L}_S\}$.

Overall, there are two different cases: (i) global versus conditional distribution, and (ii) conditional versus conditional distribution. For the first case occurring in intra-class variants of KSA and CvMA, we provide Fig. 4 for illustration. The circles together with the arrows indicate the assignments of leakage measurements, e.g., measurement $\mathcal{L}_1$ is assigned to the subset L_2 and measurement $\mathcal{L}_3$ is assigned to the subset L_1 during the partition. Subscript # is added to highlight the interpolated measurement values, i.e., those that do not actually exist. Note that the first two measurement values of L_1 are manually set to $-\infty$ to guarantee a zero CDF, since no previous values are available. The sequences of empirical CDF value pairs are marked with pink and blue, and their growth pattern follows this rule: the CDF value of the global distribution L always increases by an increment of $1/S$ at each step. This is because L is the superset of L_1 and L_2 , and contains all the actual leakage measurements. Yet, the CDF values of the conditional distributions only increase by $1/M$ or $1/N$ when the current actual measurement belongs to their specific subsets. Given that $1/M, 1/N \geq 1/S$, the two CDF values in a pair will remain close if the growth pattern is regular, e.g., the CDF value of L_2 (L_1) increases at fixed intervals of N/S (resp. M/S) steps.

For the second case of conditional versus conditional distribution, which is the core idea of inter-class variants, we first deconstruct the CDF pairs in the global versus conditional distribution, and then recombine the two CDF sequences of the conditional distributions into a new paired sequence. Note that

CDF values of leakage measurements that do not belong to any of the subsets should be excluded, e.g., the one of $\mathcal{L}_6$ in the figure. In our example, the empirical CDF value pairs of conditional versus conditional distribution are: $\{(0, 1/N), (0, 2/N), (1/M, 2/N), (1/M, 3/N), (2/M, 3/N), (3/M, 3/N), \dots, (1, 1)\}$. Its growth pattern follows another specific rule: at each step, only one CDF value in the pair can increase (by an increment of $1/M$ or $1/N$), due to the misalignment problem discussed in Section 3.1.

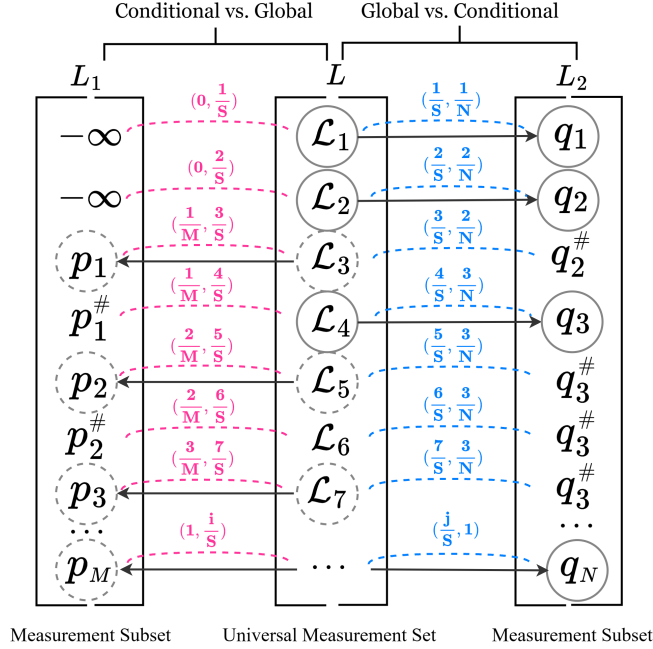


Fig. 4: The growth patter of empirical CDF value pairs.

4.2 Goodness-of-Fit Test via Ordinality and Nominality

As observed by [20] and inferred from Section 3, inter-class variants of KSA and CvMA appear to be more promising options for key recovery. In this subsection, we propose a novel distinguisher to further enhance their capabilities, by introducing ordinality and nominality to manage the leakage measurements. This allows the task of identifying meaningful partitions to be converted into a goodness-of-fit test, which has been extensively studied and verified in statistics, thus offering significant benefits regarding explainability and efficiency.

This distinguisher performs three stages for key recovery:

1. *Managing leakage measurements with ordinality.* It first sorts the measurement set in ascending order as: $L = \{\mathcal{L}_1, \mathcal{L}_2, \dots, \mathcal{L}_S\}$. This stage also features

in the calculation of empirical CDF values in KSA and CvMA, but it is not a part of DoM. It standardizes the measurements to the same format which enables the distinguisher to leverage the CDF growth pattern to enhance its performance.

2. *Managing leakage measurements with nominality.* Guessing a subkey candidate and identifying the subset to which each measurement belongs. Then, the actual measurement values are replaced by nominal values “0” and “1”. In general, nominality refers to the meaning of labels (i.e., not numerically comparable) [30]. These labels are simply subset identifiers and are interchangeable. In Fig. 4, if label “0” is assigned to L_1 and “1” to L_2 , then the measurement set L is transformed into a binary sequence $\mathbf{B} = \{001011 \dots\}$ ($\mathcal{L}_6$ is excluded). This process bases on our analysis in Section 4.1 that only one CDF value in a pair increases at each step, ensuring that each measurement value receives a single label.
3. *Applying goodness-of-fit test.* The ranking and labeling in the previous two stages yield a generic non-parametric advantage. Derived from our observation about the growth pattern, the curve stays close to the reference line in the PP plot if the two CDF values increase alternately at a similar rate. Consequently, the “0” and “1” should be uniformly and randomly spread over the binary sequence $\mathbf{B}$ for incorrect subkey candidates. In other words, it should be an instance of a binomial distribution $\mathbf{B} \sim \mathcal{B}(S, 0.5)$. The probability 0.5 is associated with the binary partition. This last stage calculates a p-value through a non-parametric goodness-of-fit test, which provides valuable statistical evidence to verify the guessing candidate.

We offer Fig. 5 for intuition. The overall Gaussian mixture consists of two components with different means and variances. They correspond to encryptions with different input plaintexts. In the left figure, the Gaussian mixture is correctly decomposed under the correct guessing subkey. As can be seen, the x-axis of leakage values is divided into three regions. In the low-value region, the blue component has significantly higher probability densities, making its label “0” cluster at the beginning of the sequence $\mathbf{B}$. Then, as the red component climbs to the same level, “0” and “1” alternately appear in the middle of $\mathbf{B}$. Finally, in the high-value region, the red one dominates the distribution so “1” cluster in the sequence end. This decomposition disrupts the randomness of $\mathbf{B}$ and can be detected through the goodness-of-fit test. In contrast, in the right figure, the decomposition under an incorrect subkey candidate results in two similar components. This leads to a random sequence that are likely to pass the test.

4.3 Implementation: The Mann-Kendall Goodness-of-Fit Test

In this paper, we adopt the famous Mann-Kendall test [21] to detect potential clusters indicative of meaningful partitions. The Mann-Kendall test is a generic non-parametric statistical tool for detecting “trends” (not correlations) in sequential data without assuming a specific distribution. It is widely applied in climate studies, hydrology, and environmental sciences.

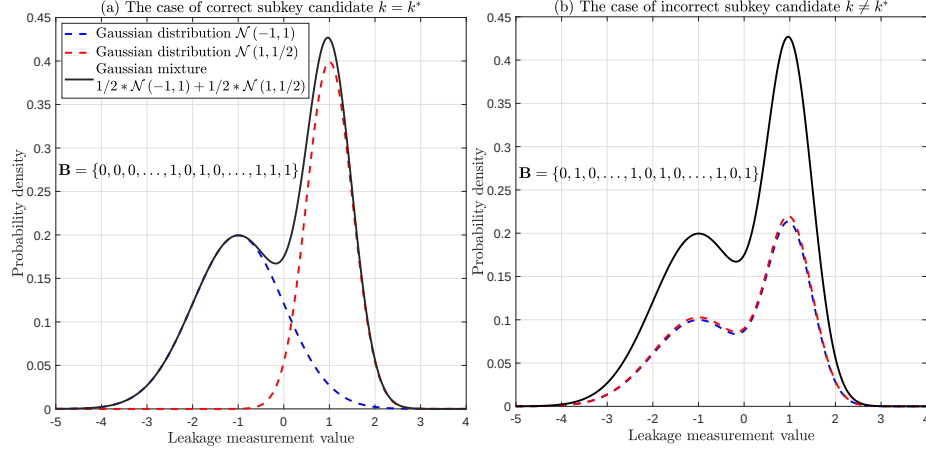


Fig. 5: The principle of goodness-of-fit test on binomial distribution.

In our scenario as a goodness-of-fit test, given the input sequence $\mathbf{B} = \{\beta_1, \beta_2, \dots\}$, the $\mathbf{S}$ statistic in the Mann-Kendall test is obtained by comparing adjacent values:

$$\mathbf{S} = \sum_{i=1} \sum_{j=i+1} \text{sgn}(\beta_j - \beta_i), \quad (16)$$

where the sign function $\text{sgn}(\cdot)$ is defined as:

$$\text{sgn}(\beta_j - \beta_i) = \begin{cases} 1 & \text{if } \beta_j > \beta_i, \\ 0 & \text{if } \beta_j = \beta_i, \\ -1 & \text{if } \beta_j < \beta_i. \end{cases} \quad (17)$$

In our particular case, the elements of sequence $\mathbf{B}$ are binary. So, the variance of $\mathbf{S}$ must be corrected to account for duplicate values (i.e., ties), for which comparisons cannot be definitively classified as positive or negative. Specifically, the adjusted variance is estimated as:

$$\text{Var}(\mathbf{S}) = \frac{n(n-1)(2n+5) - \sum_i t_i(t_i-1)(2t_i+5)}{18}. \quad (18)$$

The correction term $\sum_i t_i(t_i-1)(2t_i+5)$ is derived from all clusters of ties, where t_i represents the number of repeated values in the i -th cluster. By the central limit theorem, the standardized Z-score of the $\mathbf{S}$ statistic can now be acquired with an additional continuity correction:

$$Z = \begin{cases} \frac{S-1}{\sqrt{\text{Var}(S)}} & \text{if } S > 0, \\ 0 & \text{if } S = 0, \\ \frac{S+1}{\sqrt{\text{Var}(S)}} & \text{if } S < 0. \end{cases} \quad (19)$$

Assuming the null hypothesis H_0 states: *the sequence $\mathbf{B}$ is random and does not exhibit an upward trend (or a downward trend if the nominal labels “0” and “1” are interchanged)*, the p-value is treated as the output of this new distinguisher:

$$p_value = 1 - \Phi(Z). \quad (20)$$

Function $\Phi(z) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^z e^{-t^2/2} dt$ is the cumulative distribution function of the standard normal distribution. If “0” and “1” cluster at both ends of the sequence (e.g., as in Fig. 5(a)), this potential trend can be efficiently revealed by returning a small p_value . Experiments on simulated and real leakages in Section 6 will demonstrate the superiority of this novel distinguisher over all variants of KSA and CvMA.

5 Evaluating Moment-Based, CDF-Based and ICDF-Based Techniques

In this section, we explore the relationship between the moment-based and CDF-based techniques. We develop a criterion to guide the selection of appropriate techniques for a given context.

5.1 Bridging the Gaps

At first glance, it is not straightforward to infer the relationships directly. Therefore, we turn our analysis to the symmetry and dual counterpart of CDF in mathematics, namely the ICDF. Note that the properties of symmetry and duality do not universally apply; but they consistently hold in our side-channel scenario, as we are now illustrating:

$$\begin{aligned} \forall m \in \mathbb{F}_2^n, f(\mathbf{N}_m) > 0 &\Rightarrow \sum_{m \in \mathbb{F}_2^n} \alpha_m f(\mathbf{N}_m) > 0 \\ &\Rightarrow f(\mathcal{L}) > 0. \end{aligned}$$

$f(\cdot)$ is the probability density function (PDF) and α -s are the weight factors of Gaussian mixture (e.g., for a common uniformly distributed plaintext byte, the weights satisfy $\alpha_0 = \alpha_1 = \dots = 1/2^n$). The always positive probability density values of leakages ensure a strictly monotonically increasing CDF. Combined with its definition of being a continuous mapping, the CDF and ICDF are strictly inverse functions of each other in our side-channel scenario, demonstrating duality:

$$\begin{aligned} F_Z(F_Z^{-1}(p)) &= p, \quad p \in [0, 1] \\ F_Z^{-1}(F_Z(z)) &= z, \quad z \in \mathbb{R}. \end{aligned}$$

This bijection from real values to probabilities within $[0, 1]$ interval is measure-preserving, which guarantees a lossless transformation of information between the CDF and ICDF. In this sense, it creates a special form of symmetry: both the CDF and ICDF can perfectly restore the operations of the other, thereby maintaining the integrity of information.

Hence, we have the necessary and sufficient condition that supports our further efforts in bridging the theoretical gaps:

$$\boxed{\begin{aligned} \text{ICDF}_P \neq \text{ICDF}_Q &\Leftrightarrow \text{CDF}_P \neq \text{CDF}_Q \Leftrightarrow \text{PDF}_P \neq \text{PDF}_Q \\ &\Leftrightarrow \text{Distribution } P \neq \text{Distribution } Q. \end{aligned}}$$

5.2 The QQ Plot and ICDF-based Technique

Similar to the PP plot that relies on CDF, the QQ plot is built on ICDF. They serve the same intuitive role in comparing two distributions – examining the deviations from the reference line. But unlike the PP plot, the x-axis and y-axis turn into actual leakage measurement values. Measurements paired together have identical CDF values of their respective distributions (CDF values paired in the PP plot have identical measurement values). This extension is naturally suitable for inter-class variants, where the numbers of measurements assigned to the two partitioned subsets asymptotically approach equal. Recalling our sorted measurement subsets $L_1 = \{p_1, p_2, \dots, p_M\}$ and $L_2 = \{q_1, q_2, \dots, q_N\}$, we note that $M \approx N$. The empirical CDF sequences for both L_1 and L_2 tend to be the same: $\{1/N, 2/N, \dots, 1\} \approx \{1/M, 2/M, \dots, 1\}$. So, the discrete QQ curve results from the paired measurement sequence: $\{(p_1, q_1), (p_2, q_2), \dots\}$. Building on the CvM divergence, which has been shown to be more effective than the KS [32], we derive the ICDF-based technique expressed as:

$$\mathbf{D}_{\text{eCvMA}}^* = \arg \max_{k \in \mathcal{K}} \sum_i (p_i - q_i)^2. \quad (21)$$

Interestingly, this approach circumvents the misalignment problem without the need for any interpolation method. For traditional CDF-based techniques, the continuous real values of leakage measurements make it hard to overcome the misalignment problem. However, this task is much easier for the empirical CDF values due to their regular increasing pattern of fixed increments (refer to our analysis in Section 4.1). This interesting technique will also be validated by our experiments in Section 6. At the theoretical level, the symmetry and duality properties between the ICDF and CDF imply on par performance among the techniques employing them.

5.3 Security Evaluation

We start the evaluation by taking a look at the relationship between the CDF-based and ICDF-based techniques. Luckily, this issue has already been discussed in statistical literature (e.g., see [22]): the QQ plot is more sensitive to heavy-tailed distributions yet advantageous for complex-shaped distributions (e.g., the Gaussian mixture in this paper). In side-channel context, the Gaussian mixture assumption has been convincingly supported by the well-established Hamming weight (or distance) model. Therefore, the QQ plot may capture finer details about the distribution differences (e.g., the kurtosis in Fig. 5(a) and skewness

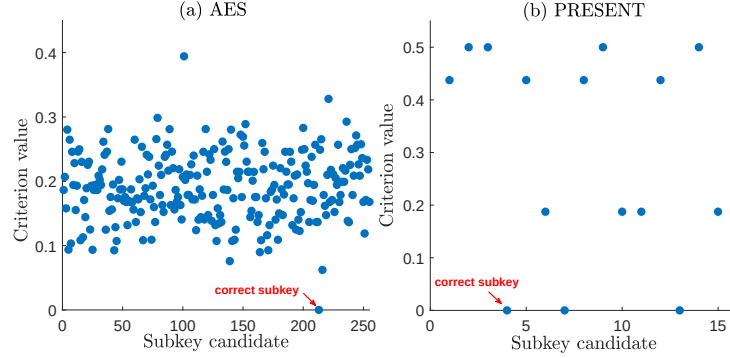


Fig. 6: The criterion value cst of AES and PRESENT S-boxes.

or multimodality when more components are involved). Low-noise condition reduces the impact of heavy-tailed behavior and enhances its effectiveness.

We then establish the relationship between the moment-based and ICDF-based techniques by expanding Equ. (21):

$$\begin{aligned}
 \mathbf{D}_{\text{eCvMA}}^* &= \arg \max_{k \in \mathcal{K}} \sum_i (p_i - q_i)^2 = \arg \max_{k \in \mathcal{K}} \hat{\mathbb{E}}\{(p_i - q_i)^2\} \\
 &= \arg \max_{k \in \mathcal{K}} \hat{\mathbb{D}}\{p_i\} + \hat{\mathbb{D}}\{q_i\} + \hat{\mathbb{E}}^2\{p_i\} + \hat{\mathbb{E}}^2\{q_i\} - 2\hat{\mathbb{E}}\{p_i q_i\} \\
 &= \arg \max_{k \in \mathcal{K}} \hat{\mathbb{D}}\{p_i\} + \hat{\mathbb{D}}\{q_i\} + (\hat{\mathbb{E}}\{p_i\} - \hat{\mathbb{E}}\{q_i\})^2 - 2\text{Cov}(p_i, q_i) \\
 &= \arg \max_{k \in \mathcal{K}} (\Delta_{\text{DoM}}^{(k)})^2 + \hat{\mathbb{D}}\{(p_i - q_i)\},
 \end{aligned} \tag{22}$$

where $\text{Cov}(\cdot)$ is the covariance operator. Notably, the first term $(\Delta_{\text{DoM}}^{(k)})^2$ is actually the robust extension of DoM in Equ. (3). Therefore, the only difference between the ICDF-based and moment-based techniques under the same partition strategy resides in the second term $\hat{\mathbb{D}}\{(p_i - q_i)\}$. According to the margin analysis, we take the population version $\mathbb{D}\{(p_i - q_i)\}$ – free from sampling bias – as the criterion cst to judge which kind of techniques is preferable and under what circumstances. If cst still maintains the highest for the correct subkey and has larger security margins than the first term $(\Delta_{\text{DoM}}^{(k)})^2$, we can infer that the ICDF-based techniques are at least no weaker than those moment-based ones. Otherwise, the relatively reduced margin will immediately lead to a performance degradation.

We examine the criterion value cst under the typical HW leakages for every subkey candidate in Fig. 6. We consider both AES and PRESENT S-boxes. We employ the non-cumulative partition strategy on the most significant bit as the example and discard the noise component. This allows us to separate out the effects of the algebraic properties of cryptographic algorithms. As shown in the figure, the criterion values for AES are concentrated in the range of 0.1 to 0.3, while those for PRESENT are dispersed. Note that the criterion values for the

correct subkeys stay at the minimum value of zero. This is because each pair of measurement values drawn from the two sorted subsets differs by a constant Hamming weight of one, attributable to the bit selection function. Therefore, the variance $\mathbb{D}\{(p_i - q_i)\}$ remains zero all the time. The criterion values will differ when targeting various bits, yet consistently remain zero for the correct subkey candidate. Similar analyses can be applied for other settings.

These observations are independent of the noise level (since we have removed its impacts), and we can draw an important conclusion that the ICDF-based techniques are generally weaker than the moment-based family for most common leakages approximating a HW model, due to the reduced margins caused by the minimal zero *cst* value for the correct subkey candidate.

Thanks to the bridge of the ICDF-based techniques, we can finally summarize the relationship between the moment-based and CDF-based techniques under the same partition strategy. For the common side-channel leakages that approximate the HW (or HD) model, we have:

$$\text{Moment-based} \geq \text{ICDF-based} \approx \text{CDF-based}. \quad (23)$$

This inequality allows for bounding security levels of the CDF-based techniques with well-studied evaluation metrics that are originally proposed for evaluating the moment-based DoM. In the following experiments, we will also show that our Mann-Kendall distinguisher is close to optimality (i.e., achieves very similar performance as DoM – its theoretical upper bound).

6 Experimental Analyses

In this section, we provide practical and detailed validation for our newly proposed side-channel distinguishers and theoretical conclusions. We investigate both the AES ($n = 8$) and PRESENT ($n = 4$) cryptographic algorithms. Real leakages of AES are from the software-implemented open dataset DPAcontest v4 [1] and hardware-implemented open dataset TeSCASE [2]. Real leakages of PRESENT are sampled from an ATMega328p micro-controller whose clock frequency is 16 MHz. We apply a WaveRunner 8104 oscilloscope with a sampling rate of 1 GS/s and collect 160,000 leakage measurements. We also conduct extensive simulations so the noise level can be well-controlled to encompass a wide range of practical scenarios, thereby demonstrating more comprehensive and convincing evaluations.

6.1 Simulated Experiments on AES and PRESENT

We consider the typical HW leakages $\mathcal{L} = \text{HW}(\text{Sbox}(m \oplus k^*)) + \mathbf{N}$ with representative SNR levels ranging from 0.01 to 1. Experimental results are displayed in Figs. 7~10. All the inter-class, intra-class, cumulative and non-cumulative variants are involved for validation. Our observations are:

1. The CvMA distinguishers based on the sum of squared deviations exhibit significantly better performance than the KSA, across all SNR levels, partition strategies, cryptographic algorithms, and numbers of measurements. These observations agree with the experimental results reported in [32], and highlight the core advantage of CvMA: its focus on the robust global deviation rather than merely the maximum position.
2. The inter-class variants of CvMA can slightly outperform their intra-class counterparts (see Figs. 9~10), as evidenced by our margin analyses in Section 3.3, which is also consistent with the simulation findings in [20]. However, for KSA, its inter-class improvements appear to be somewhat suboptimal, especially under a low SNR. This is because the maximum deviation emphasized by the KS divergence is more vulnerable to noise. Hence, taking the expectation of KS values in the intra-class variants helps mitigate the noise impacts.
3. The ICDF-based distinguisher $\mathbf{D}_{\text{eCvMA}}^*$ we suggest in Equ. (21) presents very similar behavior to that of the conventional CDF-based $\mathbf{D}_{\text{eCvMA}}$ under most experimental settings (even for the low-SNR settings in Fig. 9(b) and 10(b) where the heavy-tailed distributions may be regarded as less favorable). In Fig. 9(a) and 10(a), the success rates of $\mathbf{D}_{\text{eCvMA}}^*$ drop slightly below those of $\mathbf{D}_{\text{eCvMA}}$. This is due to the fact that only a small number of measurements is required in these settings (i.e., about 300), so noticeable variation exists in the sizes of the two partitioned subsets, and some measurements must be discarded to ensure pairing. These robust phenomena verify the symmetry and duality between CDF and ICDF in a side-channel context, as discussed in Section 5.1. It also confirms the role of this new distinguisher for being the bridge to connect the relationships between moment-based and CDF-based techniques.
4. Our derived inequality in Equ. (23) (i.e., the relationships among moment-based, ICDF-based and CDF-based techniques) holds true in all settings. For the left-hand side of the inequality, as can be seen from the figures, the moment-based DoM takes the lead all the time, regardless of the SNR levels, partition strategies, and cryptographic algorithms. This advantage is even more evident for PRESENT, which can be explained by looking at Fig. 6(b) and recalling our analyses in Section 5.3. That is, the criterion values for incorrect subkey candidates can be higher (i.e., > 0.4) than those in AES (i.e., $0.1 \sim 0.3$), therefore resulting in a greater reduction of security margins and a severe degradation in the performance of the CDF-based techniques. The close behaviors of $\mathbf{D}_{\text{eCvMA}}^*$ and $\mathbf{D}_{\text{eCvMA}}$ in the above observation validates the right-hand side of the inequality.
5. Unlike the trivial improvements made by the inter-class strategy, our novel distinguisher incorporating the famous Mann-Kendall test in statistics demonstrates impressive gains in performance. Despite being bounded by the inequality of Equ. (23) (not defeating the moment-based DoM but achieving optimality with a very close performance), this new distinguisher brings substantial enhancements to the CDF-based family. The performance gains are

robust and persist across all experimental settings, indicating that it is always a significantly better alternative to all variants of KSA and CvMA.

6.2 Real Experiments on AES – Software Implementation

The real leakages are from the popular open dataset DPAcontest v4. Detailed descriptions about this dataset are omitted for brevity. Experimental results are given in Fig. 11. The target intermediate variable x is chosen as the output of the first S-box in the first round encryption. Our observations are:

1. Different from the simulated experiments, all the CvMA variants markedly surpass the KSA family under the noncumulative partition, but only modestly under the cumulative strategy. This is because the noise level is fixed for the real leakages, and the cumulative partition operated on multiple bits of the intermediate variable tends to result in a higher SNR. This variant strategy alleviates the noise vulnerability of the KS divergence and allows its performance to keep up with the more robust CvMA method. Even so, the CvM divergence still appears to be a better choice in practice.
2. None of the inter-class variants shows a clear advantage over their intra-class competitors on these real leakages of practical scenarios, irrespective of the partition strategies, and raising the concerns that the existing inter-class improvements may be of limited effectiveness. This observation is in line with the theoretical findings in [20]: the inter-class and intra-class variants are equivalent when the leakages are roughly unimodal — a condition that holds for this dataset (see the analysis in [20] about the unimodality of the leakage distributions of the ATMega 163 microcontroller in DPAcontest v4). In contrast, the newly proposed Mann-Kendall distinguisher sustains its superiority in all settings. This observation again highlights the robustness of this new distinguisher, and confirms its necessity as a superior alternative to all variants of KSA and CvMA distinguishers.
3. The ICDF-based technique displays unexpectedly high efficiency in Fig. 11(a), where it exceeds all the CDF-based techniques. Then, in Fig. 11(b), it attains a similar performance. These observations validate the suitability of the ICDF-based techniques for handling the complex distribution of real leakages in practice. The symmetry and duality properties are preserved in practical scenarios, ensuring the bridging role of this technique.
4. Our theoretical findings regarding the moment-based, ICDF-based and CDF-based techniques in Equ. (23) remain valid in practical scenarios of real leakages. The experimental observations validate the soundness and practical relevance of our theory: it offers a framework for evaluating or bounding the security levels of commonly adopted side-channel techniques.

6.3 Real Experiments on AES – Hardware Implementation

The real leakages are from the well-known open dataset TeSCASE, which contains measurements collected on a Sasebo-GII board with a Xilinx Virtex-5

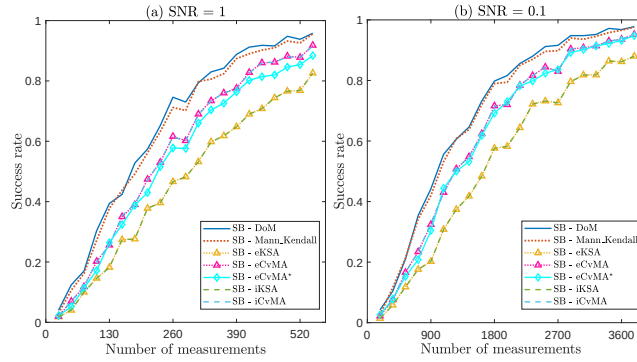


Fig. 7: Experiments on simulated HW leakages of AES – noncumulative partition.

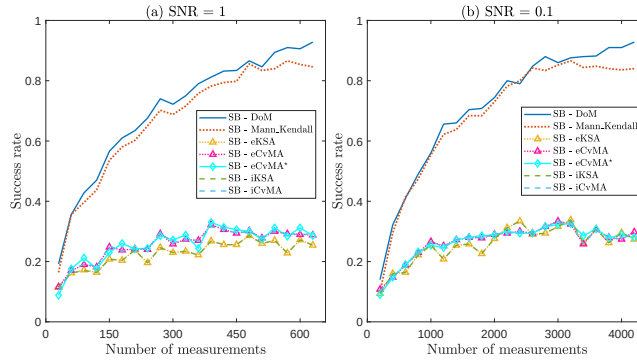


Fig. 8: Experiments on simulated HW leakages of PRESENT – noncumulative partition.

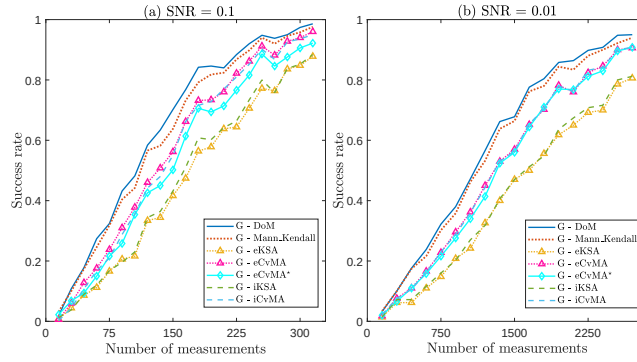


Fig. 9: Experiments on simulated HW leakages of AES – cumulative partition.

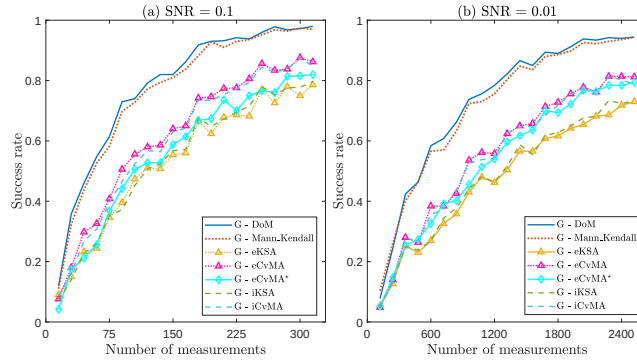


Fig. 10: Experiments on simulated HW leakages of PRESENT – cumulative partition.

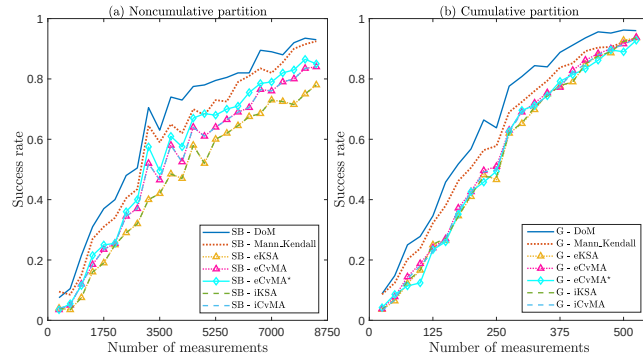


Fig. 11: Experiments on real leakages of DPAcontest v4.

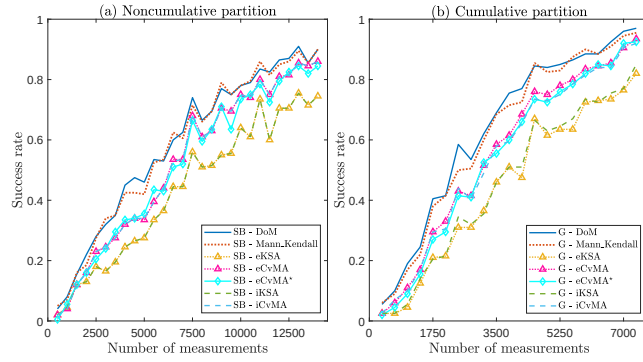


Fig. 12: Experiments on real leakages of TeSCASE.

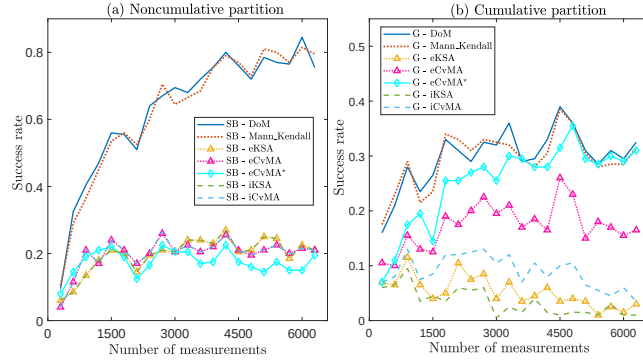


Fig. 13: Experiments on real leakages of ATMega328p.

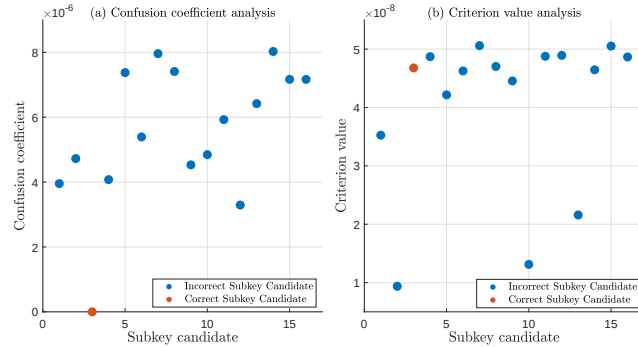


Fig. 14: Confusion coefficient and criterion value analyses.

FPGA. This dataset provides sufficient traces to support our repeated experiments, and the experimental results are given in Fig. 12. The target intermediate variable x in this hardware implementation is chosen as the bitwise XOR of the input and output of the first S-box in the last round encryption. Our observations are:

1. All the CvMA variants go beyond the KSA competitors as expected, once again demonstrating the strength and robustness of the CvM divergence in practical scenarios (note that compared to the experiments of DPAcontest v4, this gain exists under both partition strategies). This observation affirms our ICDF-based technique being grounded in the more powerful CvM divergence.
2. Like in the software implementation, the existing inter-class improvements of KSA and CvMA do not deliver significant performance gains neither, and in Fig. 12(b) the inter-class KSA variants even lag behind those original intra-class versions. These observations reveal a clear need for a further substantial improvement, which is one of the main goals we achieve in this paper.
3. Not surprisingly, our novel Mann-Kendall distinguisher continues to perform well under all partition strategies, and even presents a progressive trend of defeating the moment-based DoM. It is close to optimality (i.e., achieves very similar performance as DoM – its theoretical upper bound). This observation demonstrates the practical relevance of this distinguisher as a substantial improvement over the conventional KSA and CvMA.
4. The ICDF-based distinguisher $\mathbf{D}_{\text{eCvMA}}^*$ shows highly similar behaviors to the CDF-based $\mathbf{D}_{\text{eCvMA}}$. Validating this connection in practice supports our further conclusion that the statistical moment of leakages is more informative than the perspective provided by the cumulative distribution, as can be told from the figures.
5. Our theoretical conclusions extend without modification to practical scenarios of hardware implementation. The observations on both software and hardware implementations confirm that our theory provides a general guidance for evaluating side-channel techniques in practice.

6.4 Real Experiments on PRESENT

The real leakages are sampled from an ATmega328p micro-controller. The target intermediate variable x is the output of the first S-box in the first round encryption. Experimental results are given in Figs. 13~14. The Point-Of-Interest (POI) of the leakages is displayed in Fig. 15. Our observations are:

1. Similar to the simulated experiments (see Fig. 8 and 10), KSA and CvMA seem less effective on the PRESENT implementation: the rising trends of success rates are gentler than those observed under the AES. This phenomenon holds for all inter-class and intra-class variants, and the situation goes worse under the cumulative partition strategy (see Fig. 13(b)). Such

poor performance in practical scenarios highlights a pressing need for further optimization (note that the inter-class improvements still fail to surpass the original inter-class versions of KSA).

2. Our novel Mann-Kendall distinguisher effectively overcomes this bottleneck of the CDF-based family, elevating its success rates to the level of the moment-based DoM, and has the potential to yield even better results. This substantial improvement demonstrated in Fig. 13 can also be found in the simulation (see Fig. 8 and 10), and is verified across both the strategies. The Mann-Kendall distinguisher shows promise for practical applications.
3. The ICDF-based distinguisher sticks close to the CDF-based techniques under the noncumulative partition strategy, again proving their equivalence in the side-channel context. However, it presents a more powerful trend under the cumulative strategy since the large SNR mitigates the impacts of the heavy tail, as we discuss in Section 5.3.
4. In Fig. 13(b), all the distinguishers simultaneously experience a performance degradation. Therefore, to find out the reasons, we apply the confusion coefficient and the criterion value analysis for further explanations. As shown in Fig 14(a), all the confusion coefficients are of an extremely small magnitude (i.e., 10^{-6}), which implies great difficulty for distinguishing the correct subkey from the other candidates. This explains the poor performance of DoM, and thanks to the relationships we establish in Equ. (23), the confusion coefficient analysis proposed for the DoM can now help to bound the security levels of the KSA and CvMA as well. Furthermore, the criterion value analysis in Fig 14(b) additionally explains the distorted performance of the CDF-based techniques. Although the value corresponding to the correct subkey does not equal zero like in Fig. 6(b), it is still not the highest among all the candidates. The reduced security margin causes a distortion where the success rates decrease with the number of measurements.
5. Our findings regarding the moment-based, ICDF-based and CDF-based techniques have been verified as solidly grounded in practice, including the inferior cases where all the distinguishers exhibit unsatisfactory performance.

7 Conclusions

This paper centers on practical characteristic-aware techniques in side-channel analysis, which do not mandate a direct estimation of leakage densities. We propose to use PP plot as the framework to explain foundations of CDF-based techniques and facilitate intuitive comparisons between different variant strategies. Then, we introduce the Mann-Kendall test for advanced key recovery which is shown to be much better than all existing variants. Eventually, evaluation on relationships between moment-based and CDF-based techniques is carried out by bridging their gaps through the ICDF-based technique.

In the future, we will extend our research to other variant strategies, including non-binary select function where leakages are divided into multiple classes. We

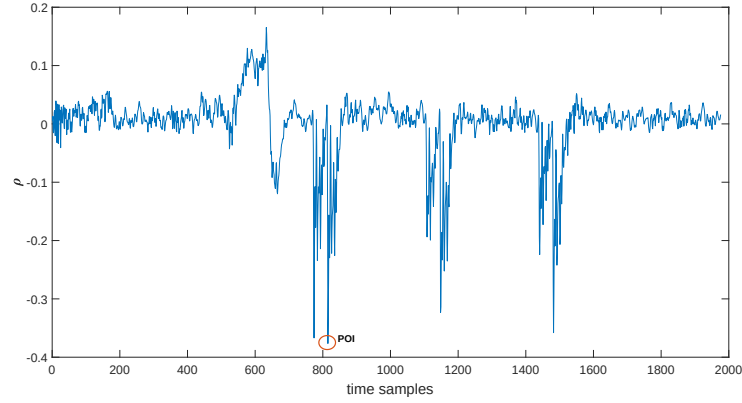


Fig. 15: The selected POI on real leakages of PRESENT on the ATmega328p micro-controller.

also plan to extend KSA and CvMA to masked scenarios which they have not been able to deal with so far (unlike DoM that can raise its moment to a higher order). We are looking forward to pinpointing the conditions under which the CDF-based and our ICDF-based techniques demonstrate unique advantages.

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